

MODULE 2

Vectors, Matrices and Linear Transformations

Mathematics for Computing

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Module 2: Overview

Main Theme

This module develops the mathematical foundations of **vectors, matrices, vector spaces and linear transformations**.

- ▶ Vectors and coordinate systems
- ▶ Linear combinations, span and basis
- ▶ Matrix multiplication and transformations
- ▶ Determinants, inverses and systems of equations
- ▶ Column space and null space
- ▶ Dot and cross products
- ▶ Change of basis
- ▶ Eigenvalues and eigenvectors
- ▶ Vector spaces

1. Vectors

Definition

A **vector** is a mathematical object that has both magnitude and direction.

- ▶ Vectors are commonly represented as ordered lists of numbers.
- ▶ A two-dimensional vector belongs to $\mathbb{R}^2$.
- ▶ A three-dimensional vector belongs to $\mathbb{R}^3$.

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} \in \mathbb{R}^n$$

Example

$$\mathbf{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

Its magnitude is

$$\|\mathbf{v}\| = \sqrt{3^2 + 4^2} = 5.$$

2. Coordinate System

A coordinate system provides a reference framework for representing points and vectors.

- ▶ In $\mathbb{R}^2$, two perpendicular axes are generally used.
- ▶ In $\mathbb{R}^3$, three mutually perpendicular axes are used.
- ▶ The origin is represented by the zero vector.

$$v = xi + yj$$

For three dimensions,

$$v = xi + yj + zk$$

where

$$i = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad j = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad k = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

3. Vector Addition

Two vectors of the same dimension can be added component by component.

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$$

Then

$$\mathbf{u} + \mathbf{v} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \\ \vdots \\ u_n + v_n \end{bmatrix}.$$

Example

$$\begin{bmatrix} 2 \\ 3 \end{bmatrix} + \begin{bmatrix} 4 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \end{bmatrix}.$$

- ▶ Vector addition is commutative: $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$.
- ▶ It is also associative.

4. Scalar Multiplication

Multiplication of a vector by a scalar changes its magnitude and, depending on the sign, its direction.

For scalar c ,

$$c\mathbf{v} = c \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = \begin{bmatrix} cv_1 \\ cv_2 \\ \vdots \\ cv_n \end{bmatrix}.$$

Example

$$3 \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 6 \\ -3 \end{bmatrix}.$$

- ▶ $c > 1$: magnitude increases.
- ▶ $0 < c < 1$: magnitude decreases.
- ▶ $c < 0$: direction is reversed.
- ▶ $c = 0$: result is the zero vector.

5. Linear Combinations

A **linear combination** is formed by multiplying vectors by scalars and adding the resulting vectors.

Given vectors

$$v_1, v_2, \dots, v_k,$$

a linear combination is

$$c_1 v_1 + c_2 v_2 + \dots + c_k v_k.$$

where $c_1, c_2, \dots, c_k$ are scalars.

Example

Let

$$v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Then

$$3v_1 + 2v_2 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}.$$

6. Span of Vectors

The **span** of a set of vectors is the collection of all possible linear combinations of those vectors.

$$\text{span}\{v_1, v_2, \dots, v_k\}$$

contains every vector that can be represented as

$$c_1v_1 + c_2v_2 + \dots + c_kv_k.$$

Geometric Interpretation

The span describes the space that can be reached using the given vectors and scalar multiplication.

- ▶ One non-zero vector in $\mathbb{R}^2$ spans a line.
- ▶ Two linearly independent vectors in $\mathbb{R}^2$ span a plane.
- ▶ Three independent vectors in $\mathbb{R}^3$ span $\mathbb{R}^3$.

7. Basis Vectors

A **basis** is a set of vectors that:

- ① spans the vector space, and
- ② is linearly independent.

The standard basis of $\mathbb{R}^2$ is

$$e_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad e_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Every vector

$$v = \begin{bmatrix} x \\ y \end{bmatrix}$$

can be written as

$$v = xe_1 + ye_2.$$

Key Idea

A basis provides a coordinate system for representing vectors.

8. Linear Independence

Vectors $v_1, \dots, v_k$ are linearly independent if

$$c_1 v_1 + \dots + c_k v_k = 0$$

has only the trivial solution

$$c_1 = c_2 = \dots = c_k = 0.$$

- ▶ Independent vectors contain no redundant direction.
- ▶ If one vector can be expressed using the others, the set is linearly dependent.
- ▶ A basis must always be linearly independent.

Example

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

are linearly independent.

9. Matrix Multiplication

For matrices A and B , the product AB is defined when

number of columns of A = number of rows of B .

If

$$A \in \mathbb{R}^{m \times n}, \quad B \in \mathbb{R}^{n \times p},$$

then

$$AB \in \mathbb{R}^{m \times p}.$$

The (i, j) entry is

$$(AB)_{ij} = \sum_{k=1}^n a_{ik} b_{kj}.$$

Important

Matrix multiplication is generally **not commutative**:

$$AB \neq BA.$$

10. Matrix Multiplication as Composition

A matrix can represent a linear transformation.

Suppose

$$T_1(x) = Ax$$

and

$$T_2(x) = Bx.$$

Applying T_1 followed by T_2 gives

$$T_2(T_1(x)) = B(Ax) = (BA)x.$$

Thus matrix multiplication represents the **composition of linear transformations**.

Interpretation

Transformation 1 $\longrightarrow$ Transformation 2 $\longrightarrow$ Combined transformation

The order of multiplication is therefore important.

11. Three-Dimensional Linear Transformations

A linear transformation maps vectors from one vector space to another while preserving vector addition and scalar multiplication.

$$T(u + v) = T(u) + T(v)$$

and

$$T(cu) = cT(u).$$

In $\mathbb{R}^3$,

$$T(x) = Ax, \quad A \in \mathbb{R}^{3 \times 3}.$$

Common transformations include:

- ▶ Rotation
- ▶ Reflection
- ▶ Scaling
- ▶ Shearing

12. Examples of 3D Linear Transformations

A scaling transformation can be represented by

$$A = \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & s_z \end{bmatrix}.$$

Then

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = A \begin{bmatrix} x \\ y \\ z \end{bmatrix}.$$

For a rotation about the z-axis,

$$R_z(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Applications

3D transformations are fundamental in:

- ▶ Computer graphics
- ▶ Computer vision
- ▶ Robotics
- ▶ Game development

13. Determinant

The **determinant** is a scalar value associated with a square matrix.

For

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

the determinant is

$$\det(A) = ad - bc.$$

Geometric Meaning

The absolute value of the determinant represents the factor by which a linear transformation changes area or volume.

- ▶ $\det(A) > 0$: orientation is preserved.
- ▶ $\det(A) < 0$: orientation is reversed.
- ▶ $\det(A) = 0$: space is collapsed into a lower dimension.

14. Determinant of a 3×3 Matrix

For

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix},$$

the determinant is

$$\det(A) = a(ei - fh) - b(di - fg) + c(dh - eg).$$

Properties

- ▶ $\det(AB) = \det(A) \det(B)$.
- ▶ $\det(A^T) = \det(A)$.
- ▶ If two rows are interchanged, the sign changes.
- ▶ A matrix is singular if $\det(A) = 0$.

15. Inverse Matrix

For a square matrix A , its inverse A^{-1} satisfies

$$AA^{-1} = A^{-1}A = I.$$

The inverse exists only when

$$\det(A) \neq 0.$$

For

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Application

The inverse matrix is useful for solving systems:

$$Ax = b$$

giving

$$x = A^{-1}b.$$

16. Column Space

The **column space** of a matrix A is the span of its column vectors.

If

$$A = \begin{bmatrix} | & | & \cdots & | \\ a_1 & a_2 & \cdots & a_n \\ | & | & \cdots & | \end{bmatrix},$$

then

$$\text{Col}(A) = \text{span}\{a_1, a_2, \dots, a_n\}.$$

- ▶ The column space describes all possible outputs Ax .
- ▶ Its dimension is called the **rank** of A .
- ▶ Pivot columns provide a basis for the column space.

17. Null Space

The **null space** of A is the set of vectors x satisfying

$$Ax = 0.$$

It is written as

$$\text{Null}(A) = \{x : Ax = 0\}.$$

Interpretation

The null space contains all input vectors that are mapped to the zero vector by the transformation represented by A .

- ▶ The null space is a vector space.
- ▶ Its dimension is called the **nullity**.

18. Rank–Nullity Theorem

For a matrix

$$A \in \mathbb{R}^{m \times n},$$

the rank–nullity theorem states

$$\text{rank}(A) + \text{nullity}(A) = n$$

where n is the number of columns of A .

Meaning

- ▶ Rank measures the number of independent output directions.
- ▶ Nullity measures the number of independent directions mapped to zero.

This theorem connects the column space and null space of a matrix.

19. Non-Square Matrices

A matrix does not have to be square.

For example,

$$A \in \mathbb{R}^{2 \times 3}$$

has two rows and three columns.

Such a matrix represents a transformation

$$T : \mathbb{R}^3 \rightarrow \mathbb{R}^2.$$

For

$$x \in \mathbb{R}^3,$$

the product

$$Ax$$

produces a vector in $\mathbb{R}^2$.

Key Idea

Non-square matrices allow transformations between spaces of different dimensions.

20. Transformations Between Dimensions

Consider

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \end{bmatrix}.$$

Then

$$A : \mathbb{R}^3 \rightarrow \mathbb{R}^2.$$

For

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix},$$

we obtain

$$A\mathbf{x} = \begin{bmatrix} x + 2z \\ y + z \end{bmatrix}.$$

- ▶ Three input coordinates are transformed into two output coordinates.
- ▶ Such transformations are common in dimensionality reduction and data representation.

21. Dot Product

For

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix},$$

the dot product is

$$\mathbf{u} \cdot \mathbf{v} = \sum_{i=1}^n u_i v_i.$$

For three dimensions,

$$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2 + u_3 v_3.$$

Geometrically,

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta.$$

22. Applications of the Dot Product

The dot product can be used to determine the angle between vectors.

$$\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}.$$

Orthogonality

Two non-zero vectors are perpendicular when

$$\mathbf{u} \cdot \mathbf{v} = 0.$$

Other applications include:

- ▶ Projection of vectors
- ▶ Distance calculations
- ▶ Computer graphics
- ▶ Machine learning similarity measures
- ▶ Physics and mechanics

23. Duality

The dot product creates a connection between vectors and linear functions.

For a fixed vector v , define

$$f(x) = v^T x.$$

This function maps

$$\mathbb{R}^n \rightarrow \mathbb{R}.$$

Thus a vector can define a linear functional.

Concept

The correspondence between vectors and linear functionals is an important idea behind **duality** in linear algebra.

In matrix notation,

$$v^T x = v_1 x_1 + \cdots + v_n x_n.$$

24. Cross Product

The cross product is defined for vectors in $\mathbb{R}^3$.

For

$$\mathbf{u} = (u_1, u_2, u_3)$$

and

$$\mathbf{v} = (v_1, v_2, v_3),$$

$$\mathbf{u} \times \mathbf{v} = \begin{bmatrix} u_2 v_3 - u_3 v_2 \\ u_3 v_1 - u_1 v_3 \\ u_1 v_2 - u_2 v_1 \end{bmatrix}.$$

The result is a vector perpendicular to both $\mathbf{u}$ and $\mathbf{v}$.

25. Properties of Cross Product

The cross product satisfies

$$\mathbf{u} \times \mathbf{v} = -(\mathbf{v} \times \mathbf{u}).$$

Therefore, it is not commutative.

Also,

$$\mathbf{u} \times \mathbf{u} = \mathbf{0}.$$

Its magnitude is

$$\|\mathbf{u} \times \mathbf{v}\| = \|\mathbf{u}\| \|\mathbf{v}\| \sin \theta.$$

Geometric Meaning

The magnitude represents the area of the parallelogram formed by $\mathbf{u}$ and $\mathbf{v}$.

26. Cross Product and Linear Transformations

The cross product can be expressed using a skew-symmetric matrix.

For

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix},$$

define

$$[\mathbf{u}]_{\times} = \begin{bmatrix} 0 & -u_3 & u_2 \\ u_3 & 0 & -u_1 \\ -u_2 & u_1 & 0 \end{bmatrix}.$$

Then

$$\boxed{\mathbf{u} \times \mathbf{v} = [\mathbf{u}]_{\times} \mathbf{v}}$$

This converts the cross-product operation into matrix multiplication.

27. Cramer's Rule

Cramer's rule provides a method for solving a system of linear equations using determinants.
For

$$Ax = b,$$

where A is an invertible $n \times n$ matrix,

$$x_i = \frac{\det(A_i)}{\det(A)}.$$

Here A_i is obtained by replacing the i -th column of A with the vector b .

Condition

Cramer's rule can be applied only when

$$\det(A) \neq 0.$$

28. Example of Cramer's Rule

Consider

$$2x + y = 5$$

$$x - y = 1.$$

The coefficient matrix is

$$A = \begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix}.$$

Therefore,

$$\det(A) = (2)(-1) - (1)(1) = -3.$$

For x ,

$$A_x = \begin{bmatrix} 5 & 1 \\ 1 & -1 \end{bmatrix}, \quad x = \frac{\det(A_x)}{\det(A)}.$$

Similarly,

$$y = \frac{\det(A_y)}{\det(A)}.$$

Thus Cramer's rule converts the solution of equations into determinant calculations.

29. Change of Basis

A vector can have different coordinate representations depending on the chosen basis.

Suppose

$$B = \{b_1, b_2, \dots, b_n\}$$

is a basis.

Then

$$x = c_1 b_1 + c_2 b_2 + \dots + c_n b_n.$$

The coordinate vector relative to basis B is

$$[x]_B = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}.$$

Important

The geometric vector remains the same, but its coordinate description changes with the basis.

30. Change-of-Basis Matrix

Let

$$P = \begin{bmatrix} | & | & \cdots & | \\ \mathbf{b}_1 & \mathbf{b}_2 & \cdots & \mathbf{b}_n \\ | & | & \cdots & | \end{bmatrix}.$$

Then

$$\mathbf{x} = P[\mathbf{x}]_B.$$

Therefore,

$$[\mathbf{x}]_B = P^{-1}\mathbf{x}.$$

For two bases B and C ,

$$[\mathbf{x}]_C = P_{C \leftarrow B}[\mathbf{x}]_B.$$

Applications

Change of basis is important in:

- ▶ Computer graphics
- ▶ Robotics
- ▶ Physics
- ▶ Coordinate transformations

31. Eigenvectors

A non-zero vector v is an **eigenvector** of matrix A if

$$Av = \lambda v.$$

Here λ is a scalar called the eigenvalue.

Interpretation

An eigenvector is a direction that does not change direction when the linear transformation A is applied to it. Only its magnitude and possibly its orientation may change.

$$Av = \lambda v.$$

32. Eigenvalues

From

$$Av = \lambda v,$$

we obtain

$$Av - \lambda v = 0.$$

Therefore,

$$(A - \lambda I)v = 0.$$

For a non-zero solution,

$$\det(A - \lambda I) = 0$$

This equation is called the **characteristic equation**.

Its roots are the eigenvalues of A .

33. Finding Eigenvectors

After finding an eigenvalue λ , substitute it into

$$(A - \lambda I)v = 0.$$

Then solve the resulting homogeneous system.

- 1 Find $\det(A - \lambda I)$.
- 2 Solve the characteristic equation.
- 3 Obtain the eigenvalues $\lambda_1, \lambda_2, \dots$
- 4 For each eigenvalue, solve

$$(A - \lambda I)v = 0.$$

- 5 The non-zero solutions are eigenvectors.

34. Example: Eigenvalues and Eigenvectors

Consider

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}.$$

The characteristic equation is

$$\det \begin{bmatrix} 2 - \lambda & 0 \\ 0 & 3 - \lambda \end{bmatrix} = 0.$$

Therefore,

$$(2 - \lambda)(3 - \lambda) = 0.$$

Hence,

$$\lambda_1 = 2, \quad \lambda_2 = 3.$$

Corresponding eigenvectors are

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

35. Vector Spaces

A **vector space** is a set of objects called vectors together with vector addition and scalar multiplication satisfying specific axioms.

Examples include:

- ▶ $\mathbb{R}^2$
- ▶ $\mathbb{R}^3$
- ▶ $\mathbb{R}^n$
- ▶ Polynomial spaces
- ▶ Spaces of matrices
- ▶ Function spaces

A vector space must be closed under:

$$u + v$$

and

$$cu.$$

36. Important Properties of Vector Spaces

For vectors u, v and scalars a, b :

- ▶ $u + v$ belongs to the space.
- ▶ au belongs to the space.
- ▶ Addition is commutative.
- ▶ Addition is associative.
- ▶ A zero vector exists.
- ▶ Every vector has an additive inverse.
- ▶ Scalar multiplication is compatible with vector addition.

Why Vector Spaces Matter

Vector spaces provide a common mathematical framework for studying vectors, matrices, transformations, data and computational models.

37. Subspaces

A **subspace** is a subset of a vector space that is itself a vector space under the same operations.

A non-empty subset W of V is a subspace if

$$u, v \in W \Rightarrow u + v \in W$$

and

$$u \in W, c \in \mathbb{R} \Rightarrow cu \in W.$$

Examples include:

- ▶ Lines through the origin
- ▶ Planes through the origin
- ▶ Column spaces
- ▶ Null spaces

38. Important Concepts at a Glance

Concept	Main Idea
Vector	Quantity represented by magnitude and direction
Linear combination	Weighted sum of vectors
Span	Set of all possible linear combinations
Basis	Linearly independent spanning set
Matrix	Representation of a linear transformation
Determinant	Measures scaling and invertibility
Column space	Span of the columns
Null space	Solutions of $Ax = 0$
Dot product	Measures alignment between vectors
Cross product	Produces a perpendicular vector in $\mathbb{R}^3$
Change of basis	Converts coordinates between bases
Eigenvector	Direction preserved by a transformation
Eigenvalue	Scaling factor associated with an eigenvector
Vector space	Set closed under vector operations

39. Applications in Computer Science

- ▶ **Computer Graphics:** transformations, rotations, scaling and coordinate systems.
- ▶ **Machine Learning:** vectors, matrices, projections, eigenvalues and eigenvectors.
- ▶ **Computer Vision:** matrix transformations and coordinate changes.
- ▶ **Data Science:** vector representations, covariance matrices and dimensionality reduction.
- ▶ **Robotics:** coordinate transformations and three-dimensional transformations.
- ▶ **Artificial Intelligence:** high-dimensional vector spaces and matrix-based computations.

40. Conclusion

Module Summary

Module 2 provides the mathematical foundation required to understand vectors, matrices, transformations and vector spaces.

- ▶ Vectors provide a representation of quantities and data.
- ▶ Linear combinations, span and basis describe vector spaces.
- ▶ Matrices represent and combine linear transformations.
- ▶ Determinants and inverses help analyze transformations and solve systems.
- ▶ Dot and cross products provide geometric relationships.
- ▶ Eigenvalues and eigenvectors reveal important directions and scaling properties.
- ▶ Vector spaces provide the general mathematical framework.

Thank You

Thank You